Shallow water equations

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Shallow water equations

- The shallow-water equations describe a thin layer of fluid of constant density in **hydrostatic balance**, bounded from below by the bottom topography and from above by a free surface.

- **Hydrostatic balance** is the state when the gravitational attraction on a fluid is exactly balanced by the pressure pushing it up.

- These equations are useful for the following reasons:

  i. They are a simpler set of equations than the full three-dimensional equations of motion, and so allow for a much more straightforward analysis of sometimes complex problems.
  ii. Despite the simplicity, the equations provide a realistic representation of a variety of phenomena in atmospheric and oceanic dynamics.

- As the fluid density is constant, the motion of the fluid can be fully determined using the momentum equations and the mass continuity equations. Due to the small scale of this, the hydrostratic balance approximation is satisfied.

Shallow water equations (SWE)

- The SWE are derived from the Navier-Stokes Equations [next slide] and are extensively used in a variety of modelling applications including:
  - Tide modelling
  - Storm surges
  - Hurricane modelling
  - Atmospheric flow modelling

- Due to the complexity of solving the SWE, the Finite Difference, Element and Volume methods have been particularly useful in solving the shallow water equations.

- The frictionless SWE are as follows:

- This is a set of three equations to determine the three variables u (east-west velocity), v (north-south velocity) and ζ (zeta, the height variations).

$$\frac{\partial u}{\partial t} + u.\frac{\partial u}{\partial x} + v.\frac{\partial u}{\partial y} = fv - g\frac{\partial \zeta}{\partial x}$$

$$\frac{\partial v}{\partial t} + u.\frac{\partial v}{\partial x} + v.\frac{\partial v}{\partial y} = -fu - g\frac{\partial \zeta}{\partial y}$$

$$\frac{\partial \zeta}{\partial t} = -\frac{\partial}{\partial x}[u(H+\zeta)] - \frac{\partial}{\partial y}[v(H+\zeta)]$$

Notes on Deriving SWE from Navier-Stokes Equations

Recall the Navier-Stokes Equations:

- Continuity Equation [mass conservation]:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

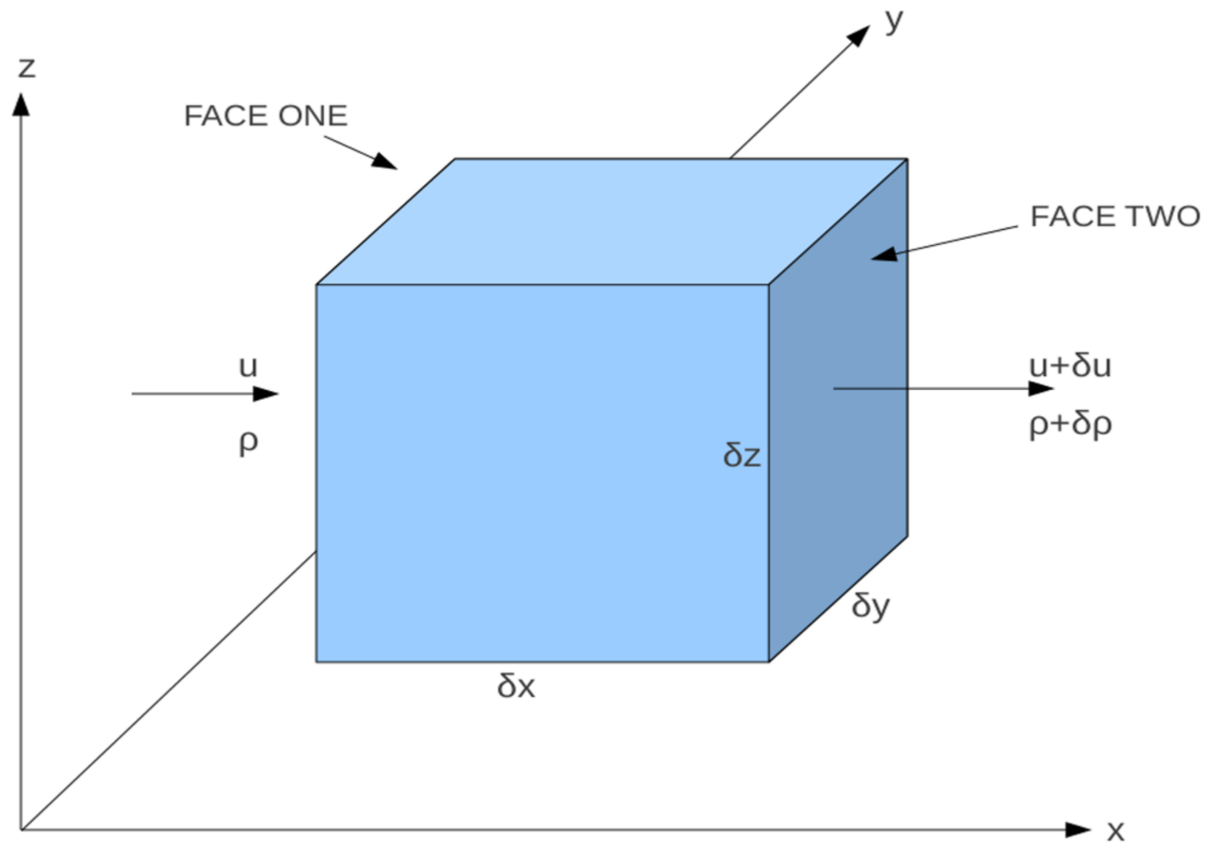
- Momentum Equation:

$$\frac{Du}{Dt} = \frac{1}{\rho} - \nabla p + v\nabla^2 u + g$$

- Hydrostatic Balance Approximation:

$$\frac{\partial p}{\partial z} = -\rho g$$

Continuity Equation (Recap)



$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

The continuity equation is the conservation of mass; i.e. what ever flows into a system must flow out of the system or accumulate within the system.

In modelling:

- It ensures model stability and accuracy
- It helps guide the development of algorithms required to develop numerical models [like the shallow water equations].
- It enforces the principle of conservation of specific properties within models.

Momentum Equation (Recap)

$$\frac{Du}{Dt} = \frac{1}{\rho} - \nabla p + v\nabla^2 u + g$$

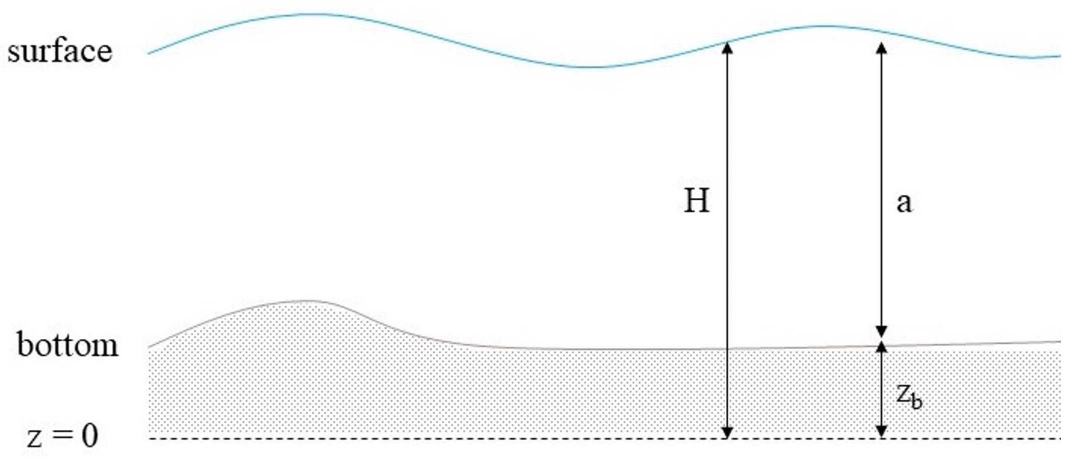
- This is a fundamental principle of fluid dynamics that is derived from the Newtons second law of motion.

- It essentially states that the rate of change of momentum of a fluid element is equal to the sum of the forces acting on it.

Important for Deriving SWE from Navier-Stokes Equations

- We also make the assumption in the SWE that we can neglect vertical acceleration as we make the assumption that the vertical scales H << than the horizontal scales.

- Before we are able to solve the SWE we need to take into account the boundary conditions, based on the surface and bottom conditions.



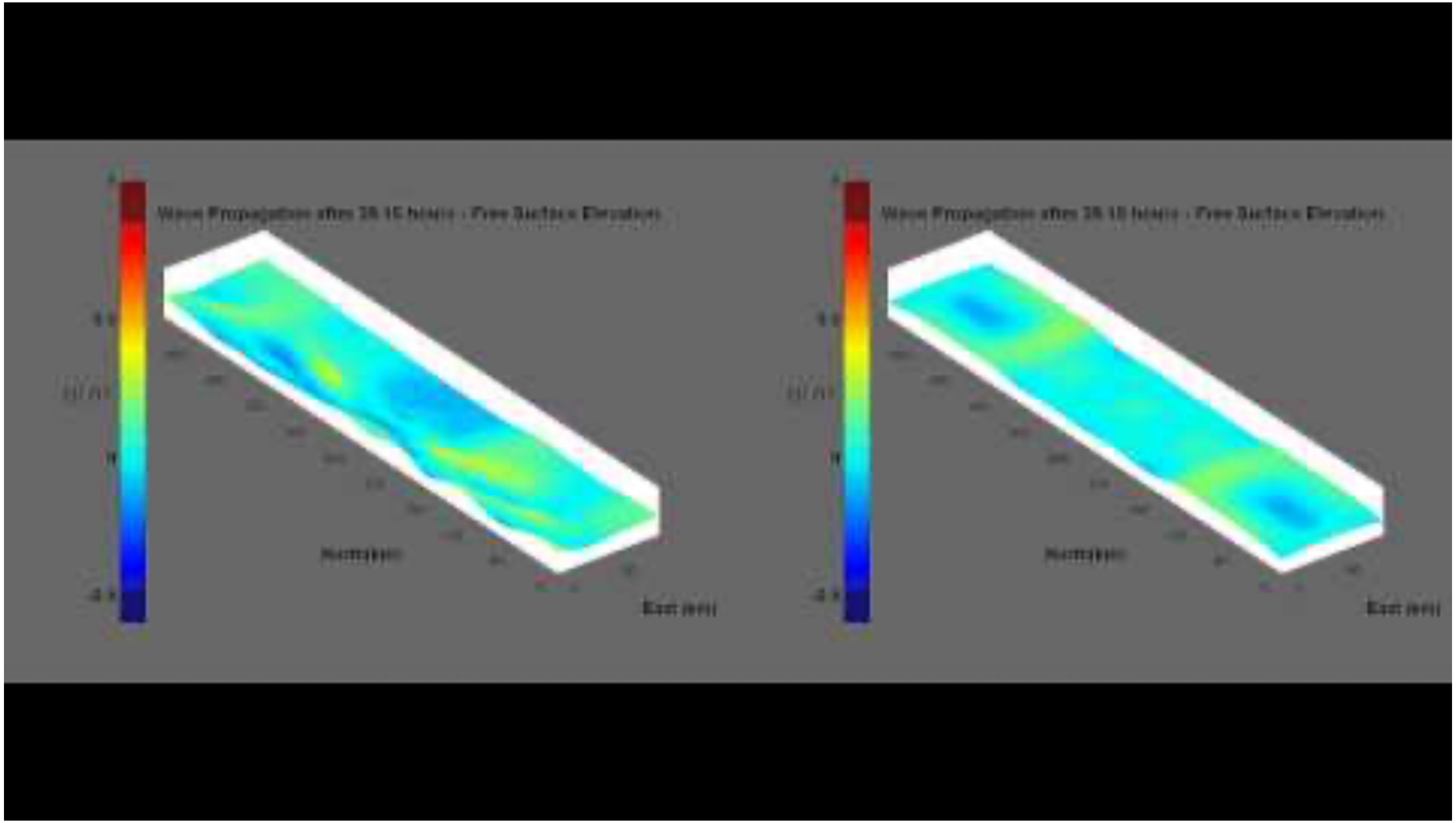
- We can consider two types of boundaries, **kinematic** and **dynamic** systems.

- Depending on whether we add an atmospheric forcing or not [we do in our practical], defines which type of boundary the surface is defined as.

- Should have been covered in your Introduction courses.

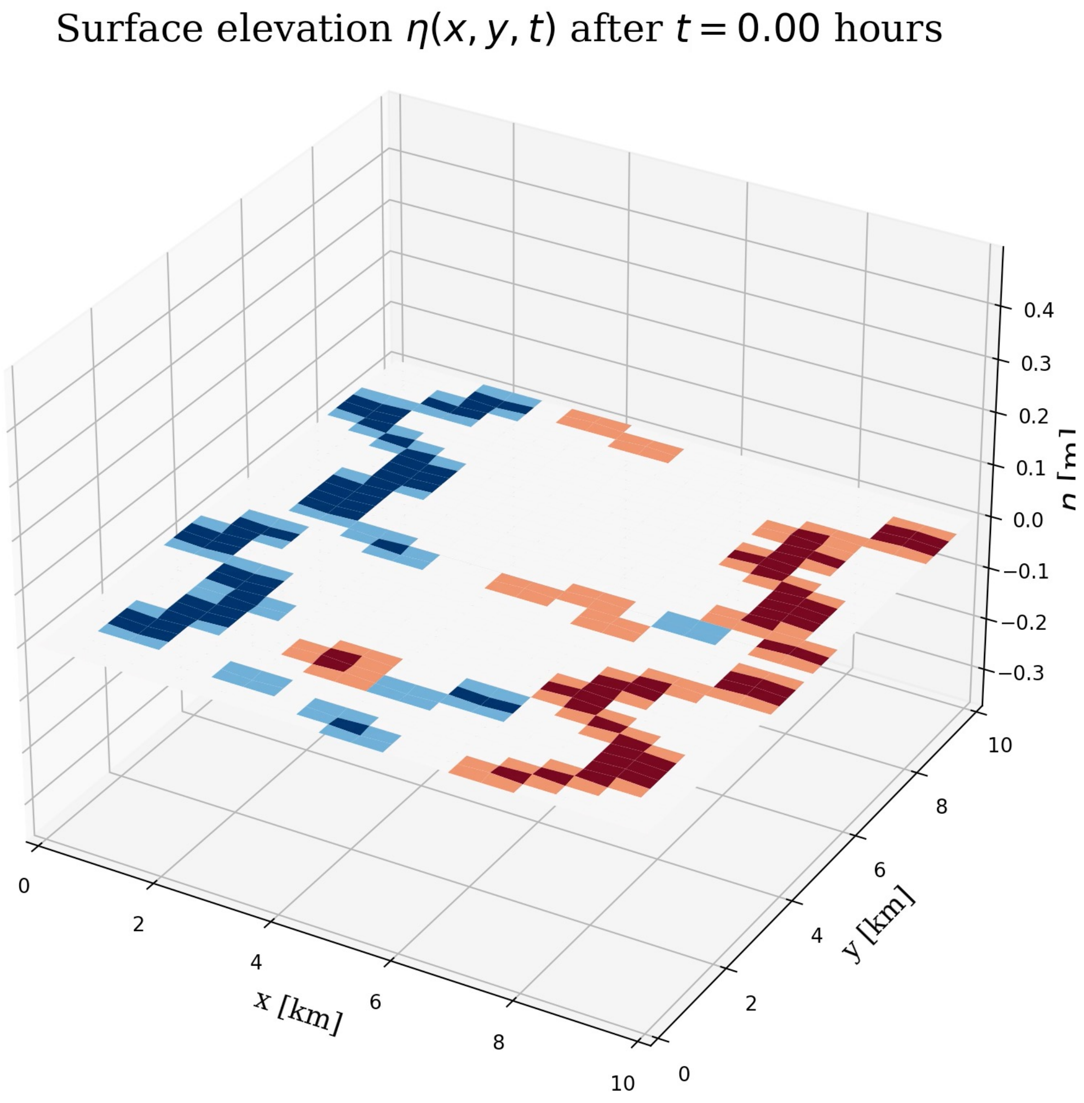
An Example

https://www.youtube.com/watch?v=AxDQWPOVzEw

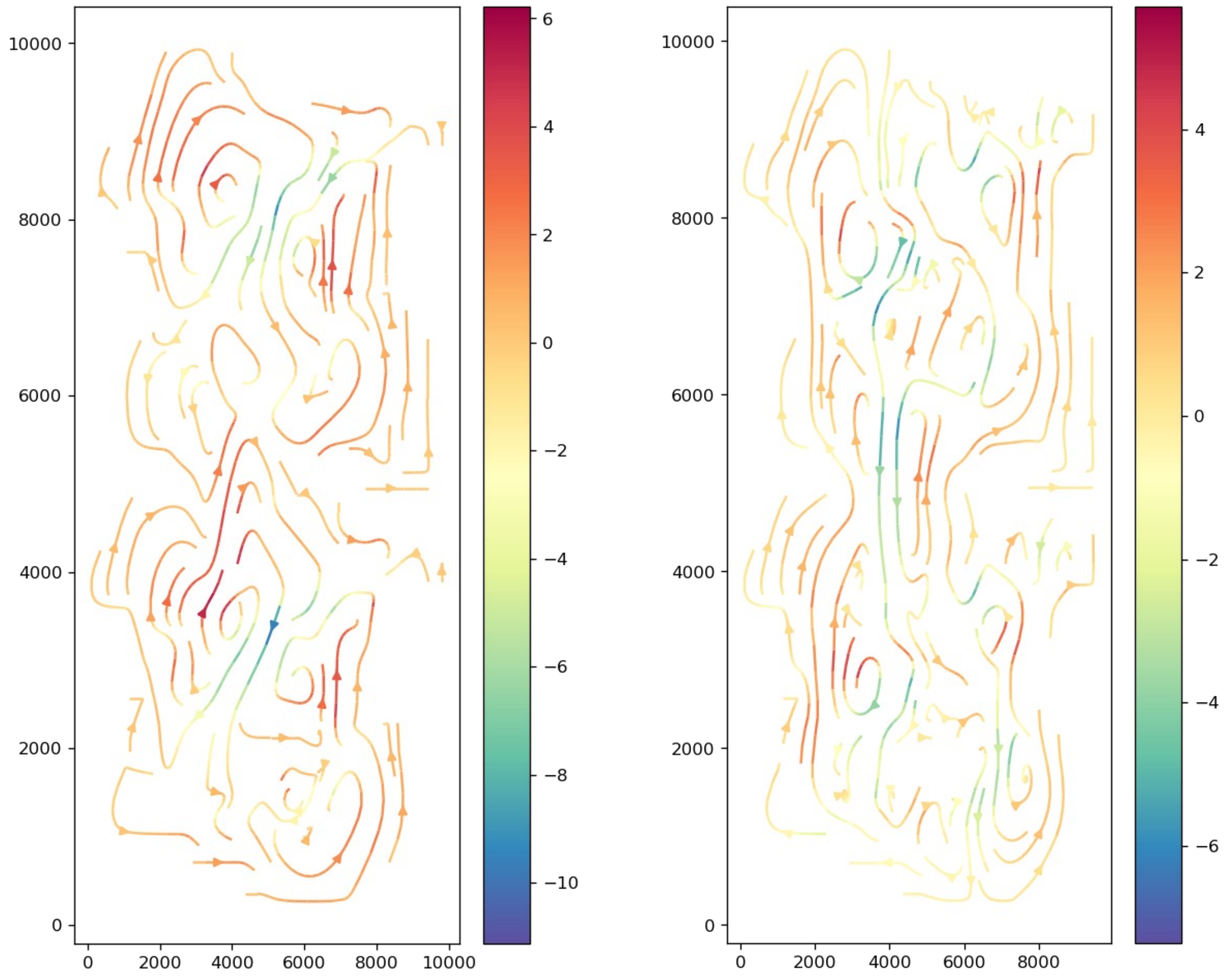


The Semester Project: Model of a Lake

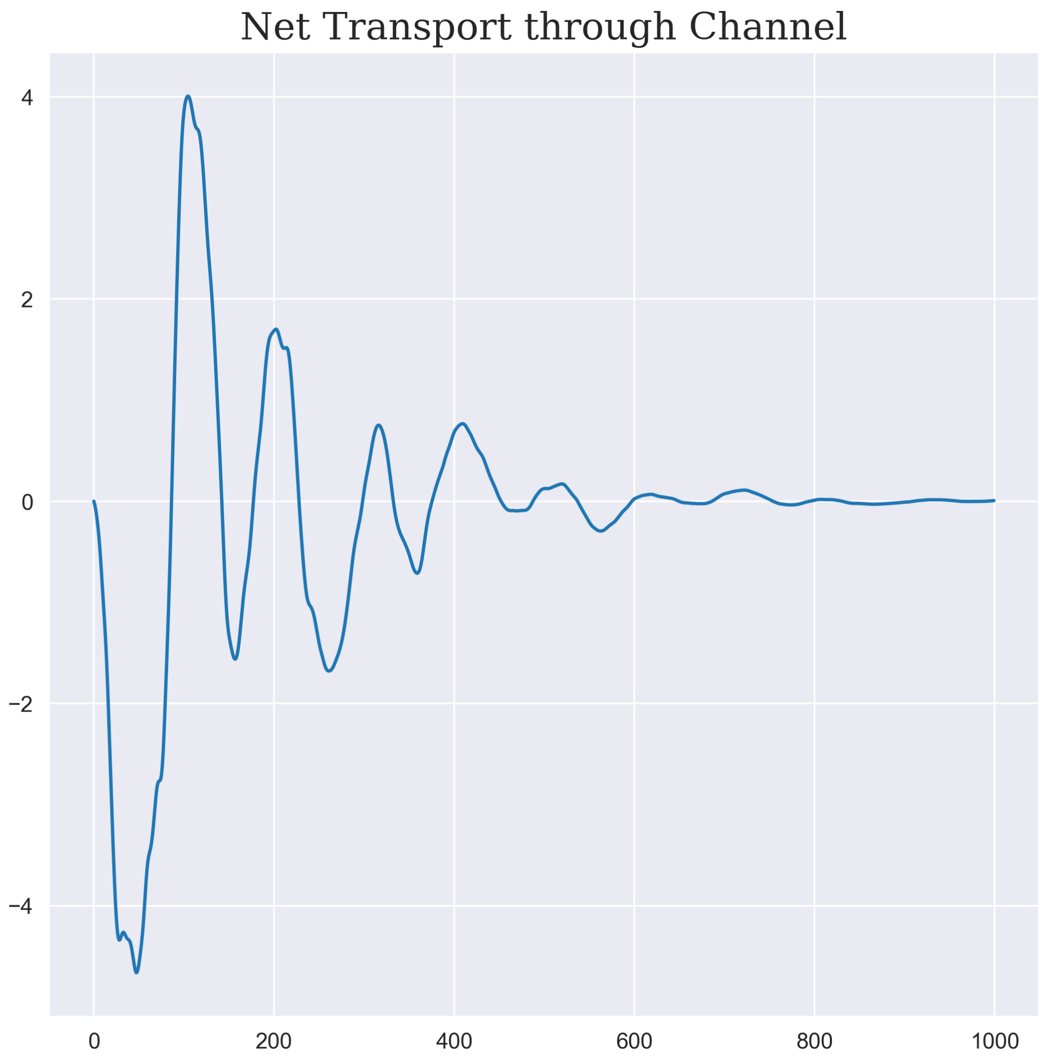
A constant 10 m/s wind!



The Semester Project: Model of a Lake



Velocity snapshots!



The Semester Project

- I will provide you a topographic map of a lake.
- The map has a maximum depth of 40 meters, with 40 by 20 grid structure.
- Let H = H(x,y) be the bottom topography (which I provide to you).
- Let $\zeta = \zeta$(x,y,t) be the sea surface height), with $u = u$(x,y,t) and $v = v$(x,y,t) be the current velocity in the east-west and north-south direction. We also have WX and WY which is constant winds which we are applying to our domain.

The Semester Project

- The linearized shallow water equations are:

$$U_t + gH\zeta_x + rU\frac{\sqrt{U^2+V^2}}{H^2} = \lambda WX\sqrt{WX^2+WY^2} + fV \qquad (1)$$

$$V_t + gH\zeta_y + rV\frac{\sqrt{U^2+V^2}}{H^2} = \lambda WY\sqrt{WX^2+WY^2} - fU \qquad (2)$$

$$\zeta_t + U_x + V_y = 0 \qquad (3)$$

- Where we have a frictional coefficient r = 0.003 and a wind stress factor of $\lambda = 3.2\times10^{-6}$.

- Here, $U = \int_{-H}^{0} udz \ and \ V = \int_{-H}^{0} vdz$ are the depth integrated transport components (units $m^2.s^{-1}$.

The Semester Project

- The linearized shallow water equations are:

$$U_t + gH\zeta_x + rU\frac{\sqrt{U^2+V^2}}{H^2} = \lambda WX\sqrt{WX^2+WY^2} + fV \tag{1}$$

$$V_t + gH\zeta_y + rV\frac{\sqrt{U^2+V^2}}{H^2} = \lambda WY\sqrt{WX^2+WY^2} - fU \tag{2}$$

$$\zeta_t + U_x + V_y = 0 \tag{3}$$

- The system we are presenting here and resolving, are influenced by a frictional drag (r), wind forcing (WX, WY and λ), Coriolis effects (f) and pressure gradients (gHζ).

- As we have Coriolis, we need a latitude, so we will be looking at 60°N.

The Semester Project: Model of a Lake

- You will produce 4 different model scenarios that will be contrasted. Run your model in each case for 1,000 time steps.

- **I will provide the four different scenarios in the next lecture!**

The Semester Project: Model of a Lake

- You will then provide me your code each scenario, so I can run the code myself in whatever langauge you decide.

- As you are giving a presentation, you will have free range to present how you want! BUT I will provide you 5 questions that you must output from your model and provide me with your code and in your presentation.

- **These 5 questions will also be given in the next lecture!**

Project Submission

- The code for the projects are due on the **8th of July for me to check!**

- **You will need to present in groups of two! Please send me an email with your pairs and I will document this and check this. If you don’t send me this by the 27th of April, we will do this directly in the class!**

- You will give a joint presentation (maximum 10 minutes) describing what you have done, how you have done, some results and animations (if you want) and any more information you want to provide yourself!

- If you provide you code and answer the questions appropriately and correctly, you will pass the assignment. You are welcome to do as much as you like!